

Transitivity $\wedge$ Independence $\wedge$ Archimedean $\Rightarrow$ Mixture continuity

Conjecture — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.
- **Archimedean.** For all $p, q, r \in \Delta(X)$: if $p \succ q \succ r$ then there are $\alpha, \beta \in (0,1)$ with $\alpha p + (1-\alpha)r \succ q$ and $q \succ \beta p + (1-\beta)r$.

0.2 Conclusion

- **Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \succcurlyeq r\}$ and $\{\lambda \in [0,1] : r \succcurlyeq \lambda p + (1-\lambda)q\}$ are closed.

0.3 Notes

Known with completeness added (Jensen 1967). Without it the cone $K = \{q - p : q \succcurlyeq p\}$ is convex; the question is whether the Archimedean axiom forces it to be closed.

Record: topics/decision-theory/results/archimedean-independence-imply-continuity.yaml